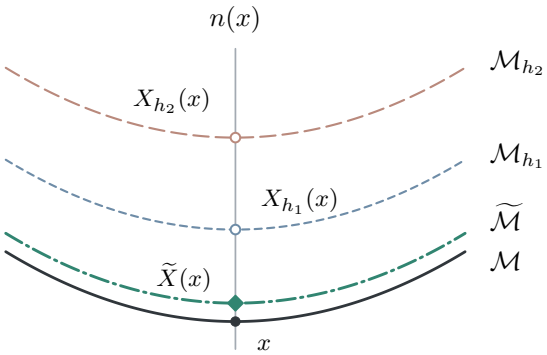
(a) Oracle-normal correspondence



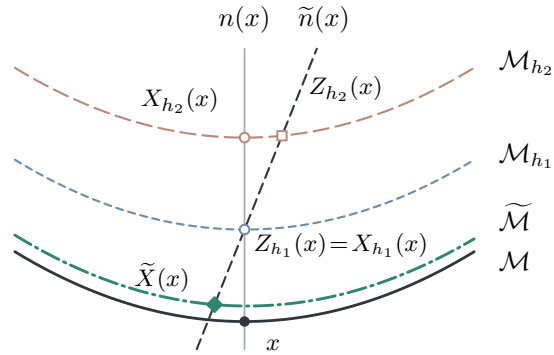

$$\widetilde{X}(x) = \sum_k w_k X_{h_k}(x)$$

$$\sum_k w_k = 1, \qquad \sum_k w_k c(h_k, \sigma) = 0$$

Cancellation of the leading normal bias

(b) Estimated-normal correspondence

$$\left\|\widetilde{n}(x) - n(x)\right\| = O(h_1^2)$$




$$\widetilde{X}(x) = \sum_k w_k Z_{h_k}(x)$$

$$\left\|\widetilde{X}(x) - x\right\| = O(h_{\max}^4)$$

*Second-order normal accuracy suffices for fourth-order correspondence.*